

F-Test and ANOVA

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How to use this guide

This guide is written for you. You'll read short explanations, then immediately run R code yourself to reproduce the results. Every step shows both the code and the output you should expect.

- By the end, you will be able to:
 - Explain what ANOVA tests and when to use it.
 - Fit a one-way ANOVA in R and interpret the output.
 - Check model assumptions with plots and simple tests.
 - Run and interpret post-hoc comparisons (Tukey HSD and adjusted t-tests).
 - Understand the idea of two-way ANOVA (factor + blocks, interaction).

Setup

You only need base R for everything in this guide. One optional test uses the `car` package.

```
# Optional: run once if you want to use Levene's test  
# install.packages("car")
```

If you use RStudio: open a new R script or Quarto document, and copy/paste the code chunks as you go.

Origins of the F-test

- William Sealy Gosset (pen name “Student”) developed the small-sample t-test while working at Guinness to improve quality control.
- Ronald A. Fisher formalised variance partitioning and introduced the F-statistic for comparing mean square quantities.

- The F-distribution is associated with Fisher and popularised by Snedecor (hence the letter “F”).
- Big idea: compare explained variation to unexplained variation using a ratio of mean squares.

From t to F: why a ratio?

- If a model explains real structure, the “between” mean square should exceed the “within” mean square.
- Under a true null, both mean squares estimate the same variance → their ratio should be near 1.
- With normal errors, these scaled sums of squares follow chi-squared distributions; their ratio follows an F-distribution with appropriate degrees of freedom.

Where the F-test appears

- One-way and multi-factor ANOVA global tests (“are any means different?”).
- Regression: overall model significance test.
- Standalone variance comparison: test whether two normal populations have equal variances.

Standalone F-test example: comparing two variances

We’ll simulate two normal samples with different variances and test $\sigma_x^2 = \sigma_y^2$.

```
set.seed(42)
x <- rnorm(30, mean = 0, sd = 1.0)
y <- rnorm(30, mean = 0, sd = 1.6)
var.test(x, y) # F-test for equality of variances
```

F test to compare two variances

```
data: x and y
F = 0.55801, num df = 29, denom df = 29, p-value = 0.122
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.2655929 1.1723758
sample estimates:
ratio of variances
```

0.5580096

How to interpret:

- H_0 : the population variances are equal. A small p-value suggests different variances.
- This test is sensitive to non-normality; use with caution if distributions are heavy-tailed or skewed.

What problem does ANOVA solve?

When you compare more than two group means (e.g., yields under fertilisers A, B, C), running many t-tests inflates the chance of a false positive (Type I error). ANOVA answers one global question:

- H_0 : $\mu_1 = \mu_2 = \dots = \mu_k$ (all group means equal)
- H_1 : at least one mean differs

It uses a single F-statistic to compare “between-groups” variation to “within-groups” variation.

From variation to sums of squares

- Total variation is partitioned into:
 - Between-groups (explained by the factor/treatment)
 - Within-groups (unexplained residual/error)
- Notation:
 - SS_T = total sum of squares
 - SS_B = between-groups sum of squares (treatment)
 - SS_W = within-groups sum of squares (residual/error)

We convert sums of squares to mean squares by dividing by degrees of freedom:

- $MS_B = SS_B / (k - 1)$, where (k) is the number of groups
- $MS_W = SS_W / (N - k)$, where (N) is the total number of observations

The F-statistic is the ratio: $F = \frac{MS_B}{MS_W}$. If all group means are truly equal, the ratio tends to be near 1; larger values suggest group means differ.

Hands-on: One-way ANOVA in R

We'll create a small dataset with one numeric response (`yield`) and one factor (`fert`).

```
fert <- data.frame(  
  yield = c(20, 22, 19, 21,  
            34, 35, 36, 34,  
            27, 28, 29, 28),  
  fert = factor(rep(c("A","B","C"), each = 4))  
)
```

Take a quick look:

```
head(fert)
```

	yield	fert
1	20	A
2	22	A
3	19	A
4	21	A
5	34	B
6	35	B

```
summary(fert)
```

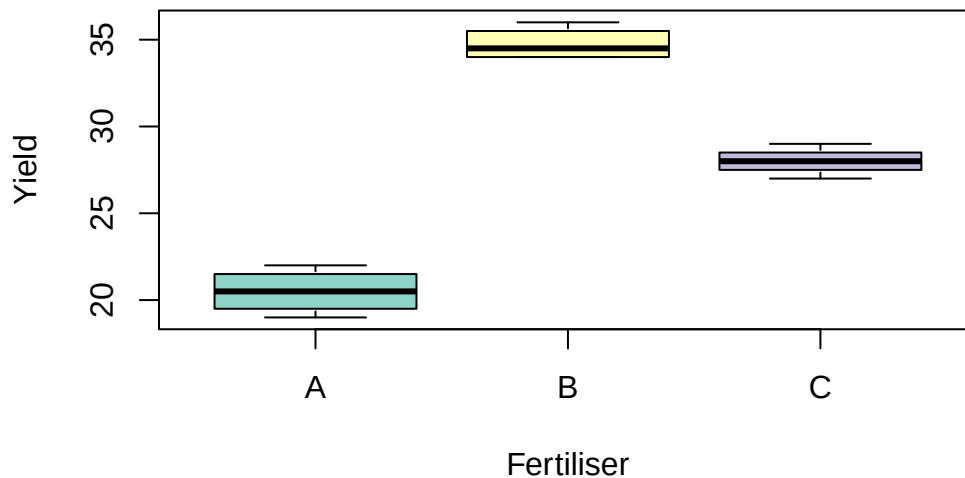
	yield	fert
Min.	:19.00	A:4
1st Qu.	:21.75	B:4
Median	:28.00	C:4
Mean	:27.75	
3rd Qu.	:34.00	
Max.	:36.00	

```
table(fert$fert)
```

A	B	C
4	4	4

Optional: visualise group differences before modelling.

```
boxplot(yield ~ fert, data = fert,
        xlab = "Fertiliser", ylab = "Yield",
        col = c("#8dd3c7", "#ffffb3", "#bebada"))
```



Fit the ANOVA model and view the ANOVA table:

```
fit <- aov(yield ~ fert, data = fert)
summary(fit)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
fert	2	406.5	203.25	187.6	4.61e-08 ***
Residuals	9	9.7	1.08		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

How to read the table:

- Df: degrees of freedom (treatment and residual)
- Sum Sq: SS_B and SS_W
- Mean Sq: MS_B and MS_W
- F value: the test statistic $F = \frac{MS_B}{MS_W}$

- $\Pr(>F)$: the p-value (small p suggests at least one mean differs)

Manual ANOVA (by hand) and verification

Let's compute the ANOVA components directly and confirm they match `aov()`:

```
# Group means and counts
group_means <- tapply(fert$yield, fert$fert, mean)
group_counts <- table(fert$fert)
grand_mean <- mean(fert$yield)

# Between-groups sum of squares (SSB)
SSB <- sum(group_counts * (group_means - grand_mean)^2)

# Within-groups sum of squares (SSW)
SSW <- sum(unlist(tapply(fert$yield, fert$fert, function(v) sum((v - mean(v))^2))))

# Degrees of freedom
k <- length(group_means)
N <- nrow(fert)
df_between <- k - 1
df_within <- N - k

# Mean squares and F
MSB <- SSB / df_between
MSW <- SSW / df_within
F_manual <- MSB / MSW

# Assemble a manual ANOVA table
anova_manual <- data.frame(
  Source = c("fert", "Residuals"),
  Df      = c(df_between, df_within),
  `Sum Sq` = c(SSB, SSW),
  `Mean Sq` = c(MSB, MSW),
  check.names = FALSE
)
anova_manual
```

	Source	Df	Sum Sq	Mean Sq
1	fert	2	406.50	203.250000
2	Residuals	9	9.75	1.083333

```
F_manual
```

```
[1] 187.6154
```

```
# Compare with aov()
anova(aov(yield ~ fert, data = fert))
```

Analysis of Variance Table

Response: yield

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
fert	2	406.50	203.250	187.62	4.607e-08 ***
Residuals	9	9.75	1.083		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

You should see identical sums of squares, degrees of freedom, mean squares, and F.

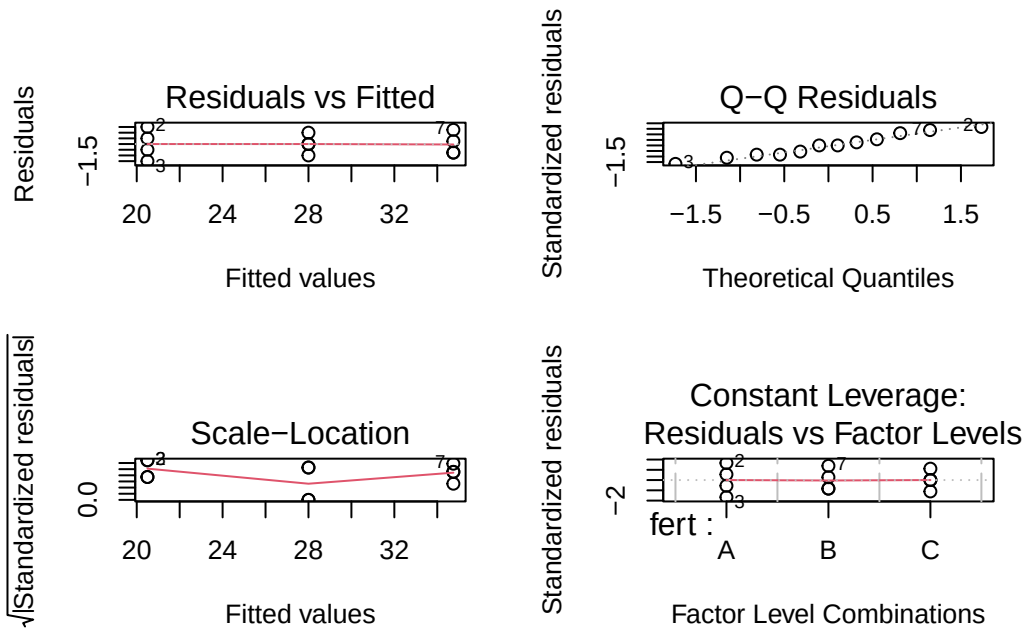
Assumptions you should check

ANOVA relies on:

1. Independence of observations (this is a design issue).
2. Approximately normal residuals.
3. Homogeneity of variances across groups.

The standard diagnostic plots are a good start:

```
par(mfrow = c(2,2))
plot(fit)
```



```
par(mfrow = c(1,1))
```

What to look for:

- Residuals vs Fitted: no strong patterns or funnels.
- Normal Q-Q: points near a straight line.

Formal checks (optional and context-dependent):

```
# Normality of residuals
shapiro.test(residuals(fit))
```

Shapiro-Wilk normality test

```
data: residuals(fit)
W = 0.96598, p-value = 0.8645
```

```
# Homogeneity of variances (choose one)
bartlett.test(yield ~ fert, data = fert) # Bartlett's test (sensitive to non-normality)
```

Bartlett test of homogeneity of variances

data: yield by fert

Bartlett's K-squared = 0.57949, df = 2, p-value = 0.7485

```
# If you installed 'car':  
# car::leveneTest(yield ~ fert, data = fert)    # Levene's test (more robust)
```

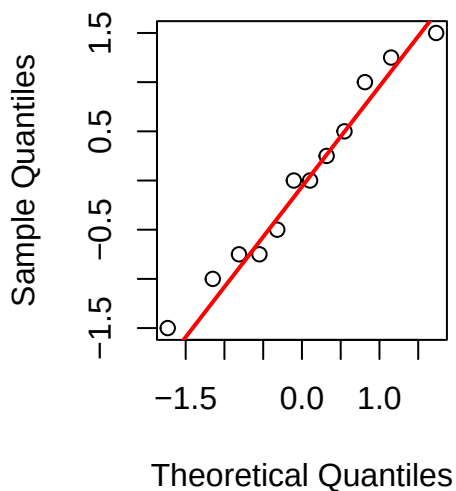
If assumptions look poor: consider transforming the response (e.g., log), using a robust method, or revisiting the design/measurement process.

Interpreting the Q-Q plot (in detail)

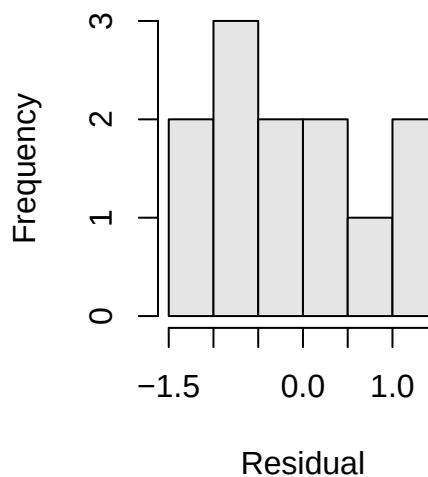
First, plot the residuals explicitly so you can inspect their shape.

```
res <- residuals(fit)  
par(mfrow = c(1,2))  
qqnorm(res, main = "Normal Q-Q Plot of Residuals")  
qqline(res, col = "red", lwd = 2)  
hist(res, breaks = 10, col = "gray90", main = "Histogram of Residuals",  
      xlab = "Residual")
```

Normal Q-Q Plot of Residuals



Histogram of Residuals



```
par(mfrow = c(1,1))
```

How to read the Q-Q plot:

- Points close to the straight line → residuals are approximately normal.
- S-shaped curve (points above the line on the left and below on the right, or vice versa) → skewness.
- Points that bow away at both ends → heavy tails (more extreme residuals than normal).
- Points that stay close in the middle but diverge strongly at ends → tail problems that can affect small-sample inference.

What to do:

- Minor deviations are usually fine; focus on large, systematic patterns.
- Consider transformations (e.g., log) for right-skewed positive data, or a different model if deviations reflect design issues.

Which means differ? Post-hoc comparisons

ANOVA tells you there is a difference somewhere. To find where, use multiple comparisons that control familywise error.

Tukey's Honest Significant Difference (HSD):

```
TukeyHSD(fit)
```

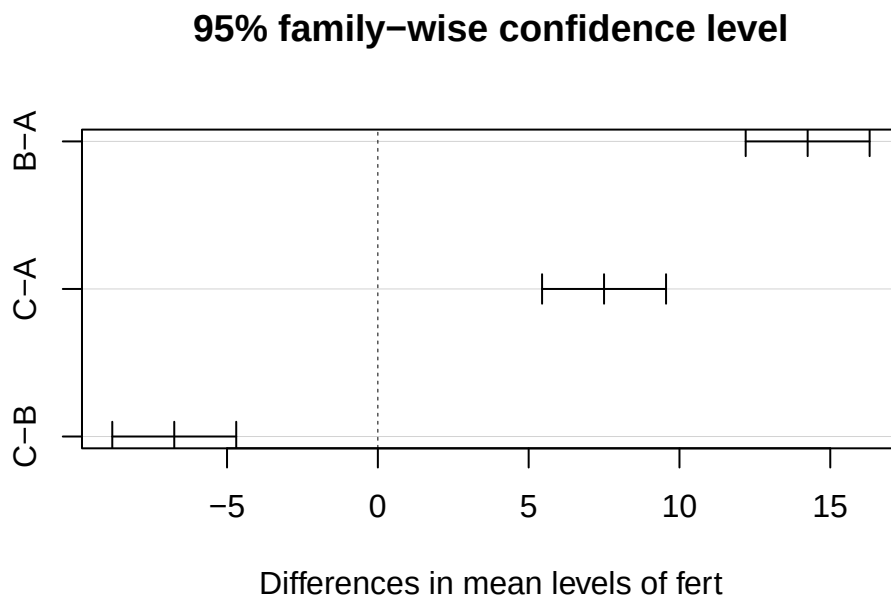
```
Tukey multiple comparisons of means
95% family-wise confidence level
```

```
Fit: aov(formula = yield ~ fert, data = fert)
```

```
$fert
      diff      lwr      upr    p adj
B-A 14.25 12.19514 16.30486 0.00e+00
C-A  7.50  5.44514  9.55486 8.20e-06
C-B -6.75 -8.80486 -4.69514 1.95e-05
```

You can also visualise the Tukey intervals:

```
plot(TukeyHSD(fit))
```



Pairwise t-tests with p-value adjustment (alternative approach):

```
pairwise.t.test(fert$yield, fert$fert, p.adjust.method = "bonferroni")
```

Pairwise comparisons using t tests with pooled SD

data: fert\$yield and fert\$fert

	A	B
B	3.6e-08	-
C	9.2e-06	2.2e-05

P value adjustment method: bonferroni

How to interpret:

- Each pair shows a difference in means, a confidence interval (Tukey), and an adjusted p-value.
- If a confidence interval does not include 0, or the adjusted p-value is small, that pair differs.

Reporting your result

Use clear, reproducible language. Replace the placeholders with your numbers:

“Yield differed by fertiliser treatment (one-way ANOVA, $F(df1, df2) = Fvalue$, $p = pvalue$). Tukey’s HSD indicated that [list groups] differed.”

Example structure: - State the ANOVA result (F , degrees of freedom, p -value). - Briefly summarise which pairs differ based on Tukey (with direction).

Extension: Two-way ANOVA (factor + blocks)

Now you’ll add a blocking factor (e.g., position) to reduce unexplained variation.

```
set.seed(1)
twoway <- expand.grid(
  fert = factor(c("A","B","C")),
  block = factor(1:4)
)

twoway$yield <- c(20, 21, 19, 22,
                 34, 33, 36, 35,
                 27, 28, 30, 29) + rnorm(12, 0, 0.6)

tw_fit <- aov(yield ~ fert + block, data = twoway)
summary(tw_fit)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
fert	2	35.24	17.62	1.11	0.3889
block	3	296.19	98.73	6.22	0.0285 *
Residuals	6	95.23	15.87		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Interpretation:

- **fert**: main treatment effect.
- **block**: accounts for position/batch effects.
- **Residual**: remaining noise.

You can also test for interaction (does the fertiliser effect depend on block?):

```
tw_fit2 <- aov(yield ~ fert * block, data = twoway)
summary(tw_fit2)
```

	Df	Sum Sq	Mean Sq
fert	2	35.24	17.62
block	3	296.19	98.73
fert:block	6	95.23	15.87

ANCOVA: ANOVA with a covariate

Why you need it:

- Sometimes groups differ partly because of a continuous baseline variable (e.g., initial size, starting weight, pre-test score).
- ANCOVA adjusts group comparisons for that covariate, improving precision and providing “fair” comparisons at the same covariate level.
- Conceptually, you fit parallel regression lines with a common slope across groups and compare their intercepts (adjusted means).

```
set.seed(2)
n_per <- 12
ancova <- data.frame(
  fert = factor(rep(c("A","B","C"), each = n_per)),
  size0 = rnorm(3 * n_per, mean = 10, sd = 2)
)
# True treatment effect and slope
beta0 <- 5; slope <- 0.8
trt_effect <- c(A = 0, B = 1.2, C = 2.0)
ancova$yield <- with(ancova, beta0 + trt_effect[fert] + slope * size0 + rnorm(nrow(ancova), 0, 1))

# Parallel slopes model (no interaction)
fit_ancova <- aov(yield ~ fert + size0, data = ancova)
summary(fit_ancova)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
fert	2	17.91	8.95	6.299	0.00494 **
size0	1	76.56	76.56	53.858	2.41e-08 ***
Residuals	32	45.49	1.42		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Optional: test for non-parallel slopes
fit_ancova_int <- aov(yield ~ fert * size0, data = ancova)
anova(fit_ancova, fit_ancova_int)
```

Analysis of Variance Table

Model 1: yield ~ fert + size0

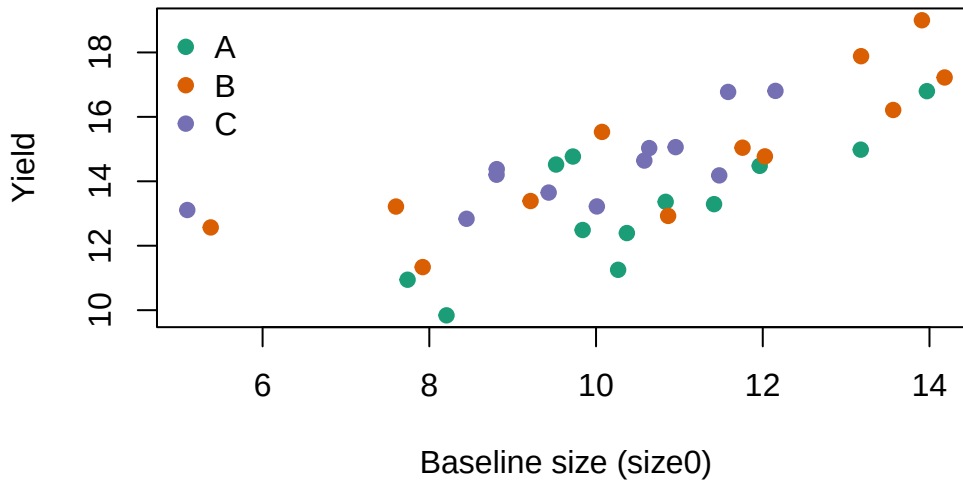
Model 2: yield ~ fert * size0

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	32	45.489				
2	30	42.782	2	2.7065	0.9489	0.3985

Visualising parallel slopes (from the ANCOVA fit):

```
lm_ancova <- lm(yield ~ fert + size0, data = ancova)
cols <- c(A = "#1b9e77", B = "#d95f02", C = "#7570b3")
plot(yield ~ size0, data = ancova, col = cols[fert], pch = 19,
     main = "ANCOVA: parallel slopes by treatment", xlab = "Baseline size (size0)", ylab = "yield")
legend("topleft", legend = levels(ancova$fert), col = cols, pch = 19, bty = "n")
slope_hat <- coef(lm_ancova)["size0"]
intercepts <- c(
  A = coef(lm_ancova)["(Intercept)"],
  B = coef(lm_ancova)["(Intercept)"] + coef(lm_ancova)["fertB"],
  C = coef(lm_ancova)["(Intercept)"] + coef(lm_ancova)["fertC"]
)
xg <- seq(min(ancova$size0), max(ancova$size0), length.out = 100)
for (g in names(intercepts)) {
  lines(xg, intercepts[g] + slope_hat * xg, col = cols[g], lwd = 2)
}
```

ANCOVA: parallel slopes by treatment



Interpretation:

- `size0` reduces residual variation by explaining baseline differences; `fert` tests adjusted group shifts.
- If the `fert:size0` interaction is not significant, parallel slopes are a good summary; if significant, the treatment effect depends on the covariate.

Assumptions (in addition to ANOVA's):

- Linearity between the covariate and the response within groups.
- Homogeneous slopes across groups (if using the parallel slopes model).
- Covariate measured without substantial error and not affected by treatment (pre-treatment).

MANOVA: multiple responses

Why you need it:

- Many experiments record multiple, correlated outcomes (e.g., plant height and leaf area; multiple biochemical markers).
- Running separate ANOVAs inflates the familywise error rate and ignores correlation structure.

- MANOVA tests for differences in the multivariate mean vector across groups, leveraging correlation to increase power and control Type I error.

What it provides:

- A single global test (e.g., Wilks' Lambda) for whether groups differ on the joint set of responses.
- If significant, you can follow up with univariate ANOVAs and multivariate visualisation to understand where differences arise.

```
set.seed(3)
m <- data.frame(
  fert = factor(rep(c("A","B","C"), each = 15))
)
mu <- cbind(A = c(0, 0), B = c(0.8, 0.4), C = c(1.2, 0.9))
Sigma <- matrix(c(1.0, 0.5, 0.5, 1.2), 2, 2) # positive correlation
# If MASS is not available, fall back to independent normals
if (!requireNamespace("MASS", quietly = TRUE)) {
  set.seed(3)
  y1 <- c(rnorm(15, mu[1, "A"], sqrt(Sigma[1,1])),
          rnorm(15, mu[1, "B"], sqrt(Sigma[1,1])),
          rnorm(15, mu[1, "C"], sqrt(Sigma[1,1])))
  y2 <- c(rnorm(15, mu[2, "A"], sqrt(Sigma[2,2])),
          rnorm(15, mu[2, "B"], sqrt(Sigma[2,2])),
          rnorm(15, mu[2, "C"], sqrt(Sigma[2,2])))
} else {
  set.seed(3)
  yA <- MASS::mvrnorm(15, mu = mu[, "A"], Sigma = Sigma)
  yB <- MASS::mvrnorm(15, mu = mu[, "B"], Sigma = Sigma)
  yC <- MASS::mvrnorm(15, mu = mu[, "C"], Sigma = Sigma)
  y1 <- c(yA[,1], yB[,1], yC[,1])
  y2 <- c(yA[,2], yB[,2], yC[,2])
}
m$y1 <- y1
m$y2 <- y2

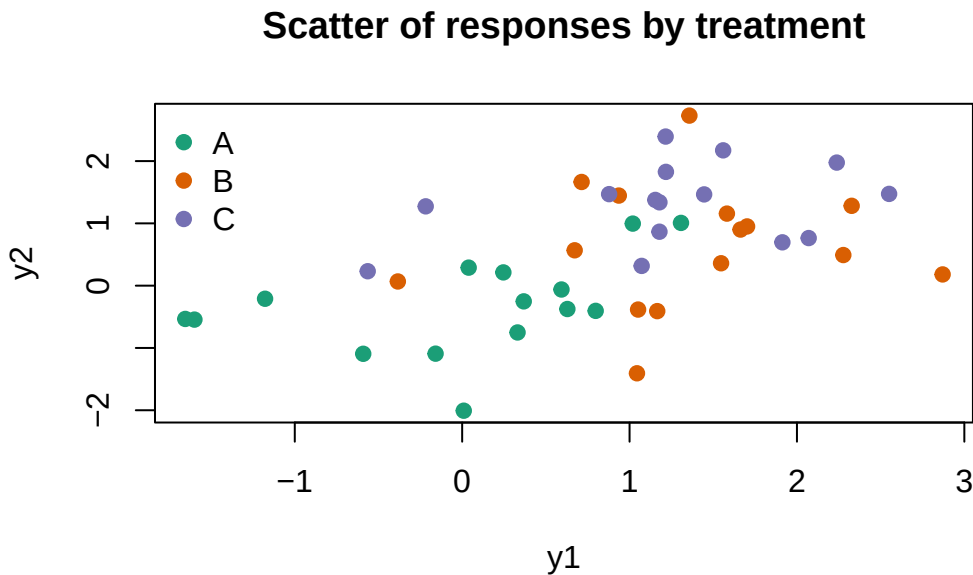
fit_man <- manova(cbind(y1, y2) ~ fert, data = m)
summary(fit_man, test = "Wilks")
```

	Df	Wilks	approx F	num Df	den Df	Pr(>F)
fert	2	0.4634	9.6144	4	82	2e-06 ***
Residuals	42					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Quick visual check of separation:

```
cols <- c(A = "#1b9e77", B = "#d95f02", C = "#7570b3")
plot(m$y1, m$y2, col = cols[m$fert], pch = 19,
      xlab = "y1", ylab = "y2", main = "Scatter of responses by treatment")
legend("topleft", legend = levels(m$fert), col = cols, pch = 19, bty = "n")
```



Interpretation and assumptions:

- A significant Wilks' test suggests group centroids differ in the 2D space.
- Assumptions: multivariate normality of residuals, equal covariance matrices across groups (Box's M), and independence. With small samples, be cautious.
- Follow-ups: inspect univariate ANOVAs and confidence ellipses (or discriminant analysis) to interpret which responses drive the separation.

Repeated measures ANOVA (within-subjects)

Why you need it:

- The same subject (or plot) is measured under multiple conditions or times; repeated measures increase efficiency but violate independence.
- Repeated measures ANOVA accounts for within-subject correlation by modelling subjects as a random block and testing the within-subject factor.

```
set.seed(4)
subjects <- factor(1:12)
rm_dat <- expand.grid(subject = subjects, treatment = factor(c("A","B","C")))
# Random subject intercepts
subj_eff <- rnorm(length(subjects), 0, 0.8)
rm_dat$response <- with(rm_dat,
  5 + (treatment == "B")*0.8 + (treatment == "C")*1.5 +
  subj_eff[subject] + rnorm(nrow(rm_dat), 0, 0.5)
)
fit_rm <- aov(response ~ treatment + Error(subject/treatment), data = rm_dat)
summary(fit_rm)
```

Error: subject

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Residuals	11	18.19	1.654		

Error: subject:treatment

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
treatment	2	8.899	4.449	21.792e-06	***
Residuals	22	4.661	0.212		

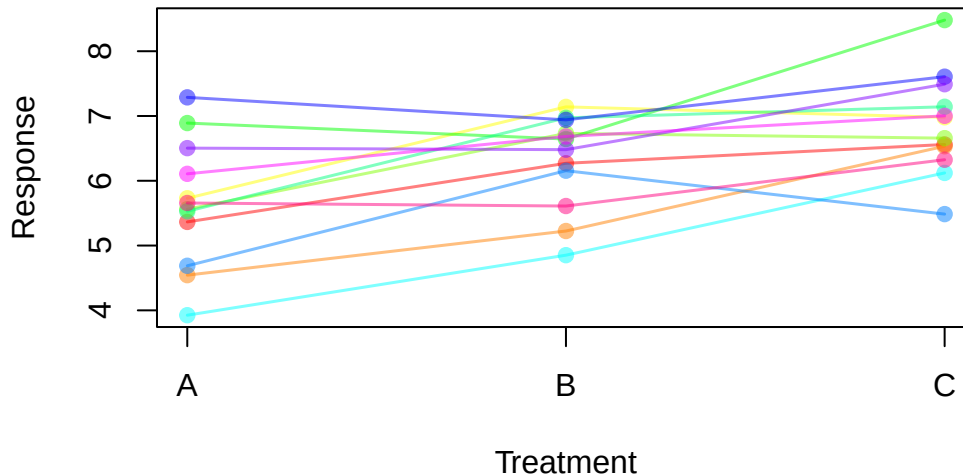
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Simple visualisation of subject trajectories:

```
cols <- rainbow(length(subjects), alpha = 0.5)
plot(1, type = "n", xlim = c(1, length(levels(rm_dat$treatment))), ylim = range(rm_dat$response),
  xaxt = "n", xlab = "Treatment", ylab = "Response", main = "Subject trajectories across t",
  axis(1, at = 1:length(levels(rm_dat$treatment)), labels = levels(rm_dat$treatment))
for (s in levels(subjects)) {
  ds <- subset(rm_dat, subject == s)
```

```
ord <- order(ds$treatment)
lines(1:length(ord), ds$response[ord], col = cols[as.integer(s)], lwd = 1.5)
points(1:length(ord), ds$response[ord], col = cols[as.integer(s)], pch = 19)
}
```

Subject trajectories across treatments



Interpretation and assumptions:

- The within-subject test of **treatment** appears under **Within** in the **Error: subject:treatment** stratum.
- Sphericity (equal variances of differences between all pairs of levels) is an additional assumption for factors with >2 levels; if violated, corrections (Greenhouse–Geisser, Huynh–Feldt) are used. In more complex designs, consider linear mixed models.

Common pitfalls and tips

- Unequal group sizes are fine, but extreme imbalance can reduce power.
- Strong outliers can inflate SS_W or distort group means—investigate them.
- With small samples, non-normal residuals can make p-values less reliable.
- Violated independence (e.g., repeated measures) calls for different models (e.g., mixed effects).

Practice: Try it yourself

1. Create your own dataset with 4 groups and 6 observations per group. Make two groups similar and two clearly different. Run the full workflow (boxplot → ANOVA → diagnostics → Tukey), and write a two-sentence report.
2. Modify the dataset so one group has a much larger variance. What happens to the diagnostics and post-hoc results?
3. Build a two-way dataset with a small block effect. Compare `aov(yield ~ group)` vs `aov(yield ~ group + block)`. How does adding blocks change the residual variation and the p-value for `group`?